

DANUJA WIJERATHNA

AI Engineer | AI Solution Architect

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years building production AI systems on Microsoft Azure. Day-to-day work covers multi-agent orchestration on Azure AI Foundry, Agentic RAG over enterprise data, MCP-based tool integration, and the LLMOps pipelines that ship these systems safely. Comfortable end-to-end: Azure Data Factory ingestion, Microsoft Fabric semantic layers, Azure AI Search hybrid retrieval, knowledge graphs, and Azure DevOps CI/CD. Increasingly stepping into solution-architect work: target architectures, build-vs-buy decisions, and design reviews on cross-team AI initiatives.

PROFESSIONAL EXPERIENCE

Senior AI Engineer

Oct 2025 – Present

IFS R&D International (Pvt) Ltd • Colombo, Sri Lanka

- Architected multi-agent AI systems on **Azure AI Foundry** using **Microsoft Agent Framework**, with specialised agents handling HR, finance, and customer-support workflows.
- Built **Agentic RAG** pipelines that combine **Graph RAG**, hybrid vector and semantic search on **Azure AI Search**, and conversation-memory management for multi-turn reasoning over enterprise data.
- Built **Model Context Protocol (MCP)** servers that expose ERP, ticketing, and HR systems as governed tools for AI agents, with audit-ready traces on every action.
- Defined the **LLMOps** reference architecture across multi-subscription Azure environments: Terraform and Bicep for IaC, automated CI/CD pipelines, model evaluation, observability, and Responsible AI guardrails for production **Generative AI** workloads.
- Built ingestion and ETL pipelines on Azure Data Factory and Azure Databricks that feed vector stores and Lakehouse layers with curated, lineage-tracked data from structured and unstructured sources.
- Built reusable semantic layers in Microsoft Fabric over large enterprise datasets so AI agents and BI consumers query consistent business metrics through one model.
- Shipped AI-powered developer tools (automated code analyser and test-script generator), cutting QA review effort by about 25 hours per week.
- Built time-series forecasting models for employee utilisation at 87% accuracy, used for resource planning across delivery teams.
- Acted as solution architect on cross-team AI initiatives: target architectures, build-vs-buy decisions, and design reviews with engineering leadership.

Senior Data Scientist

Sep 2024 – Sep 2025

IFS R&D International (Pvt) Ltd • Colombo, Sri Lanka

- Led migration of legacy Oracle databases to a modern Azure data architecture (Azure Data Factory, Azure Synapse Analytics, ADLS Gen2), with better query performance and scalability.
- Implemented a GDPR-compliant, AI-driven employee performance tracking system with row-level security using Power Apps, Power BI, and Dataverse.
- Designed and automated end-to-end ETL/ELT pipelines in Azure Data Factory and Databricks for cross-system data integration, warehousing into Synapse and Microsoft Fabric.

Data Scientist

Sep 2023 – Sep 2024

IFS R&D International (Pvt) Ltd • Colombo, Sri Lanka

- Designed semantic models in Microsoft Fabric for large enterprise datasets, which sped up Power BI reports across business units.
- Built a customer churn model using deep learning (Artificial Neural Networks) at 89% accuracy, used by account-management teams to prioritise retention.

Data Engineer

Dec 2022 – Aug 2023

AI Palette • Singapore

- Built real-time data processing pipelines and cleaned up existing ETL workflows, which cut time-to-insight for product analytics teams.
- Designed an analytics layer over the data lake that supported custom analysis and consumer-behaviour insights for downstream ML and BI consumers.
- Built ETL solutions for loading CSV and Parquet files into data lakes and cloud warehouses, used by BI and machine learning teams.

Data Science Intern (Undergraduate Trainee)

Dec 2021 – Dec 2022

IFS R&D International (Pvt) Ltd • Colombo, Sri Lanka

- Automated the Global Customer Support process, which reduced manual workload and improved response times across multi-region teams.
- Built a customer-solution prediction model in TensorFlow that anticipated customer needs and recommended proactive support actions.

KEY PROJECTS

ServiceNow MCP Server: Agentic ITSM Intelligence Platform

Stack: Model Context Protocol (MCP), ServiceNow REST APIs, Graph Database, Azure AI Foundry IQ, K-Means Clustering, Community Detection, Graph Deep Learning, NLP, Agentic AI

- Turns ServiceNow into an **AI-operable platform**: a governed **MCP** toolset over a **knowledge graph**, with **multi-hop root-cause analysis**, **K-Means + community detection** surfacing systemic issues, and **Foundry IQ** multi-index RAG grounding.

Case study: danuja-wije.github.io/#/project/servicenow-mcp

AI Power BI Report Builder: AI-Assisted Reporting Co-Working Space

Stack: Electron, React, TypeScript, Azure AI Foundry (GPT-5.4), Anthropic Claude, Model Context Protocol (MCP), Ajv

- Natural-language co-working space for Power BI: **six specialist agents** build and repair reports conversationally, with a report index cutting LLM tokens **~95%** and crash-safe, schema-validated writes.

Case study: danuja-wije.github.io/#/project/powerbi-report-builder

Project Helix: Conversational BI for Enterprise Leadership

Stack: Python 3.11, FastAPI, Claude Agent SDK (Claude Opus 4.8), in-process MCP tools, Power BI / Microsoft Fabric, live DAX, MSAL, Server-Sent Events

- Certified Power BI models answered in plain language: **live DAX**, ML insights, and a **PII-masking firewall**, governed by construction on the **Claude Agent SDK**.

Case study: danuja-wije.github.io/#/project/project-helix

Virtual Resource Manager

Stack: FastAPI (Python 3.11), React 18, TypeScript, Azure AI Foundry, Azure AI Search, Azure SQL, MSAL / Azure AD, Azure Container Apps

- Multi-agent matching engine pairing **854 employee profiles** to projects in seconds: **hybrid vector + keyword retrieval** and a **10-dimensional** configurable scoring engine.

Case study: danuja-wije.github.io/#/project/virtual-resource-manager

Log Analyzer: AI Root Cause Analysis Platform

Stack: FastAPI (Python 3.11), PostgreSQL (JSONB), Parquet, DuckDB, Azure OpenAI (GPT-5.2), Microsoft Agent Framework, Azure AI Search

- An **11-agent pipeline** distilling tens of millions of log lines per incident into executive-grade **RCA reports** with every claim cited back to the underlying logs.

Case study: danuja-wije.github.io/#/project/log-analyzer

Cloud Performance Tester (JMeter DSL Studio)

Stack: Spring Boot (Java), React, Azure OpenAI (GPT-5.2 on Azure AI Foundry), JMeter DSL

- Plain English in, performance tests out: the LLM **authors, repairs, and explains** runnable **JMeter DSL** scripts for enterprise cloud workflows.

Case study: danuja-wije.github.io/#/project/performance-tester

TECHNICAL SKILLS

AI & Generative AI: Large Language Models (LLMs), Agentic AI, Multi-Agent Orchestration, Retrieval-Augmented Generation (RAG), Graph RAG, Agentic RAG, Knowledge Graphs, Vector Embeddings, Semantic Search, Hybrid Search, Prompt Engineering, Fine-tuning, Model Context Protocol (MCP)

Azure AI Platform: Azure AI Foundry, Foundry Agent Service, Azure OpenAI Service, Azure AI Search, Azure Document Intelligence, Azure Cognitive Services, Azure Machine Learning

Agent Frameworks: Microsoft Agent Framework, Claude Agent SDK, LangChain, LangGraph, Semantic Kernel, AutoGen

Databricks: Azure Databricks, Medallion Architecture (Bronze / Silver / Gold), Delta Lake, MLflow Experiment Tracking & Model Registry, Databricks Notebooks, Databricks Jobs & Workflows, SQL Warehouses

Data Platform: Azure Data Factory (ADF), Microsoft Fabric, Azure Synapse Analytics, Azure Data Lake (ADLS Gen2), Lakehouse Architecture, Semantic Models, Power BI, Power Apps, Dataverse, PostgreSQL, DuckDB

MLOps / LLMOps: Azure DevOps, GitHub Actions, MLflow, Azure ML Prompt Flow, Terraform, Bicep, Docker, CI/CD Pipelines, Model Evaluation & Monitoring, Drift Detection, Infrastructure as Code (IaC)

Programming: Python, TypeScript, JavaScript, SQL, PySpark

Architecture & Cloud: Solution Architecture, Cloud-Native Design, Microservices, Serverless, REST APIs, Microsoft Entra ID, Zero Trust, Responsible AI Governance, GDPR Compliance, Row-Level Security

Machine Learning: Deep Learning, Artificial Neural Networks (ANN), Time Series Forecasting, Natural Language Processing (NLP), TensorFlow, PyTorch, scikit-learn, Pandas, NumPy

EDUCATION

University of Westminster, United Kingdom
MSc in Applied Artificial Intelligence

Jan 2026 – Present

Sri Lanka Institute of Information Technology (SLIIT), Malabe
BSc (Hons) in Information Technology, Specialisation: Data Science

2019 – 2023

ACHIEVEMENTS

- IFS R&D High-Performance Individual recognition (2025): selected for the company's High-Performance Team.
- IFS Aspire Leadership Programme (2025): nominated for performance-based leadership development.
- NBQSA National ICT Award (2023): Finalist, Top 5 Inter-University in Sri Lanka.
- SLASSCOM National Ingenuity Awards (2022): Best Innovative Regional Project.
- STEMUP Hack AI (2022): Finalist, Top 5 Inter-University in Sri Lanka.
- Dean's List Award (2021).
- Google Kick Start & Hash Code (2021): global rankings 13,429 and 6,661 respectively.